



Bikkad IT Institute

Why Python?

Compiled by Bikkad IT Institute | Teaching Notes

OUR LOCATIONS

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Why Python?

Why Python as a Language?

Easy to Learn

- Simple, English-like syntax
- Beginner-friendly
- Focus on logic, not complex code

Versatile

- Web development (Django, Flask)
- Automation and scripting
- Data science and AI
- Testing and DevOps

Rich Ecosystem

- 500,000+ packages on PyPI
- "Batteries included" standard library
- Strong community support

Cross-Platform & Free

- Runs on Windows, macOS, Linux
- Open source and free to use

Why Python for Data Science?

Simple Syntax

- Focus on data insights, not code complexity
- Faster prototyping
- Less boilerplate than Java/C++

Powerful Libraries

Data Manipulation:

- **pandas** - Data cleaning and analysis
- **NumPy** - Numerical computing
- **SciPy** - Scientific computing

Visualization:

- **Matplotlib, Seaborn, Plotly** - Charts and dashboards

Machine Learning:

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- **scikit-learn** - ML algorithms
- **TensorFlow, PyTorch** - Deep learning

Multiple Data Sources

- CSV, Excel, JSON, SQL databases
- Web APIs, big data tools (Spark, Hadoop)
- Cloud platforms (AWS, Azure, GCP)

Industry Standard

- Most popular for Data Analyst, Data Scientist, ML Engineer roles
- High demand and competitive salaries
- Used by Google, Netflix, Meta, Microsoft, NASA

Quick Comparison

Aspect	Python (General)	Python (Data Science)
Learning	Simple, beginner-friendly	Minimal syntax overhead
Strengths	Versatility, automation	Data wrangling, ML, visualization
Key Tools	Django, Flask, standard lib	pandas, NumPy, scikit-learn
Career	Software dev, DevOps	Data analyst, data scientist, ML

Table 1: Python General vs Data Science

Real-World Uses

- **General:** Web apps, automation, testing, system administration
- **Data Science:** Customer analysis, sales forecasting, fraud detection, recommendation systems, image recognition

Getting Started

1. Install Python from python.org (v3.11+)
2. Learn basics: variables, loops, functions
3. **For Data Science:**
 - Install Anaconda distribution
 - Learn pandas and NumPy
 - Practice with Jupyter Notebooks
 - Explore visualization (Matplotlib)
 - Move to ML (scikit-learn)

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Prepared for: Data Science & Programming Students

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